

The Geonovum OGC Checker

A linter for OGC standards

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- Geospatial data standards are part of the Dutch ‘comply or explain’ list of National standards
- One of Geonovum’s tasks is to support implementation and adoption of these standards, e.g. through
 - Dutch profiles
 - Best practice documents
 - Examples
 - Implementation in open source tooling through e.g. testbeds
 - **Validation tooling**
- At an OGC code sprint in 2023 we started to work on the “OGC checker”
 - A linter for OGC standards
 - First version created in 3 days, support for JSON-FG
 - Supports several OGC standards
 - Still under development



“We” = me and Joost Farla



The OGC checker



- Open source: <https://github.com/Geonovum-labs/ogc-checker>
- Runs as online validator at <https://geonovum-labs.github.io/ogc-checker/>
- As of now, supports validation of:

JSON-FG

OGC API Features part 1+2

OGC API Records

OGC API Processes

A screenshot of the Geonovum OGC Checker: JSON-FG web application. The interface has a dark blue header with the title 'Geonovum OGC Checker: JSON-FG', a 'URI:' label, an input field, a 'Check' button, and a dropdown menu set to 'JSON-FG'. The main area is split into two panels. The left panel shows a JSON document being validated, with line numbers 1 through 33 on the left margin. The JSON content includes fields for 'type', 'id', 'conformsTo', 'featureType', 'featureSchema', 'time', 'coordRefSys', and 'place'. The right panel displays three green status messages: '[json-fg-1/0.2/conf/core] No violations found.', '[json-fg-1/0.2/conf/3d] No violations found.', and '[json-fg-1/0.2/conf/types-schemas] No violations found.'.

```
1 {
2   "type": "Feature",
3   "id": "DENW19AL0000giv5BL",
4   "conformsTo": [
5     "[ogc-json-fg-1-0.2:core]",
6     "[ogc-json-fg-1-0.2:types-schemas]",
7     "[ogc-json-fg-1-0.2:3d]"
8   ],
9   "featureType": "app:building",
10  "featureSchema": "https://example.org/data/v1/collections/buildings/schema",
11  "time": {
12    "interval": [
13      "2014-04-24T10:50:18Z",
14      ".."
15    ]
16  },
17  "coordRefSys": "http://www.opengis.net/def/crs/EPSSG/0/5555",
18  "place": {
19    "type": "Polyhedron",
20    "coordinates": [
21      [
22        [
23          [
24            [
25              479816.67,
26              5705861.672,
27              100
28            ],
29            [
30              479822.187,
31              5705866.783,
32              100
33            ]
34          ]
35        ]
36      ]
37    ]
38  }
39 }
```



Checker for OGC API Features

What it checks:

- JSON Syntax
- Conformance to OAS 3.0
- OGC API Features part 1 v1.0.1 validation rules
- OGC API Features part 2 v1.0.1 validation rules

Requirement	Testable	Tested	Remarks
/req/crs/crs-uri	No	No	
/req/crs/fc-md-crs-list	Yes	Yes	
/req/crs/fc-md-storageCrs	No	No	
/req/crs/fc-md-storageCrs-valid-value	Yes	Yes	
/req/crs/fc-md-crs-list-global	No	No	
/req/crs/fc-bbox-crs-definition	Yes	Yes	
/req/crs/fc-bbox-crs-valid-value	Yes	Yes	Tests the presence of a 400 response.
/req/crs/fc-bbox-crs-valid-defaultValue	Yes	Yes	Covered by /req/crs/fc-bbox-crs-definition .
/req/crs/fc-bbox-crs-action	No	No	
/req/crs/fc-crs-definition	Yes	Yes	
/req/crs/fc-crs-valid-value	No	No	
/req/crs/fc-crs-default-value	Yes	Yes	Covered by /req/crs/fc-crs-definition .
/req/crs/fc-crs-action	No	No	
/req/crs/geojson	No	No	
/req/crs/ogc-crs-header	Yes	Yes	
/req/crs/ogc-crs-header-value	No	No	

Requirement	Testable	Tested	Remarks
/req/core/root-op	Yes	Yes	
/req/core/root-success	Yes	Yes	
/req/core/conformance-op	Yes	Yes	
/req/core/conformance-success	Yes	Yes	
/req/core/fc-md-op	Yes	Yes	
/req/core/fc-md-success	Yes	Yes	
/req/core/sfc-md-op	Yes	Yes	
/req/core/sfc-md-success	Yes	Yes	
/req/core/fc-op	Yes	Yes	
/req/core/fc-response	Yes	Yes	
/req/core/fc-limit-definition	Yes	Yes	
/req/core/fc-bbox-definition	Yes	Yes	
/req/core/fc-time-definition	Yes	Yes	
/req/core/f-op	Yes	Yes	
/req/core/f-response	Yes	Yes	
/req/oas30/oas-definition-2	Yes	Yes	

Problem: profile based negotiation

Table 17 – API – Records MIME Types

Encoding		Media Type	Profile Identifier
Record	HTML	text/html	n/a
	JSON	application/json	ogc-record
	GeoJSON	application/geo+json	ogc-record
Catalog	HTML	text/html	n/a
	JSON	application/json	ogc-catalog
		application/ogc-catalog+json	n/a

- Content in an OGC API is identified by media type + profile identifier
- **This means that the direct correlation between media type and OAS content schema is lost**
- This means the linter cannot differentiate the content and therefore cannot check content against its specific schema

Sprint goals

- Have a discussion on profile based negotiation with OGC API SWG members
 - Adapt linter according to outcome
- Backlog: <https://github.com/orgs/Geonovum-labs/projects/1>

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Thank you!

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